GAL Seminar 1

24 Feb 2025

Să se rezolve

ex 1

$$\begin{cases} 2x + 3y = 5 \\ 3x + 2y = 10 \end{cases}$$

ex 2

$$\begin{cases} x + y + 3z = 1 \\ 3x - y + z = 4 \\ x + 3y - 2z = 10 \end{cases}$$

ex 3

$$\begin{cases} x + y + 3z = 5 \\ 2x + 2y - 4z = 10 \\ x + y - 7z = 5 \end{cases}$$

Metoda Gauss - Jordan

ex 5

$$\begin{cases} x - y + 3z = 2 \\ 2x + 4y - 5z = 7 \\ 3x + 2y + 8z = 10 \end{cases}$$

Metoda 1 : Pivotare parțială

Metoda 2 : Pivotare totală

ex 6

$$\begin{cases} x - y - 3z = 4 \\ 3x + 3y - z = 7 \\ 5x + y - 7z = 10 \end{cases}$$

Să se rezolve

ex 1

$$\begin{cases} 2x + 3y = 5 \\ 3x + 2y = 10 \end{cases}$$

Sol:

$$\begin{cases} 2x + 3y = 5 \\ 3x + 2y = 10 \end{cases} \left| \begin{array}{l} \cdot -2 \\ \cdot 3 \end{array} \right.$$

$$\begin{cases} -4x - 6y = -10 \\ 9x + 6y = 30 \end{cases}$$

$$5x = 20$$

$$x = 4 \Rightarrow 3 \cdot 4 + 2 \cdot y = 10$$

$$2y = 10 - 12$$

$$2y = -2$$

$$y = -1$$

$\square$

ex 2

$$\begin{cases} x + y + 3z = 1 \\ 3x - y + z = 4 \\ x + 3y - 2z = 10 \end{cases}$$

Sol :

Metoda 1 : Crammer

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 3 & -1 & 1 \\ 1 & 3 & -2 \end{pmatrix}$$

$$\Delta = \begin{vmatrix} 1 & 1 & 3 \\ 3 & -1 & 1 \\ 1 & 3 & -2 \end{vmatrix} \quad \begin{array}{l} L_1 \leftarrow L_1 + L_2 \\ = \\ L_3 \leftarrow L_3 + 3L_2 \end{array}$$

$$= \begin{vmatrix} 4 & 0 & 4 \\ 3 & \boxed{-1} & 1 \\ 10 & 0 & 1 \end{vmatrix} = (-1)\cdot(-1)^{2+2} \cdot \begin{vmatrix} 4 & 4 \\ 10 & 1 \end{vmatrix}$$

$$= -(4-40) = 36$$

$$\Delta = 36$$

$$x = \frac{\Delta x}{\Delta} \qquad y = \frac{\Delta y}{\Delta} \qquad z = \frac{\Delta z}{\Delta}$$

$$\Delta x = \begin{vmatrix} \boxed{1} & 1 & 3 \\ 4 & -1 & 1 \\ 10 & 3 & -2 \end{vmatrix} \quad \begin{array}{l} C_2 \leftarrow C_2 - C_1 \\ = \\ C_3 \leftarrow C_3 - 3C_1 \end{array} \quad \begin{vmatrix} \boxed{1} & 0 & 0 \\ 4 & -5 & -11 \\ 10 & -7 & -32 \end{vmatrix}$$

$$= \begin{vmatrix} -5 & -11 \\ -7 & -32 \end{vmatrix} = 160 - 77 = 83$$

$$\Delta y = \begin{vmatrix} 1 & 1 & 3 \\ 3 & 4 & 1 \\ 1 & 10 & -2 \end{vmatrix} \quad \begin{array}{l} C_2 \leftarrow C_2 - C_1 \\ = \\ C_3 \leftarrow C_3 - 3C_1 \end{array} \quad \begin{vmatrix} 1 & 0 & 0 \\ 3 & 1 & -8 \\ 1 & 9 & -5 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -8 \\ 9 & -5 \end{vmatrix} = -5 + 72 = 67$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 1 \\ 3 & -1 & 4 \\ 1 & 3 & 10 \end{vmatrix} \begin{matrix} C_2 \leftarrow C_2 - C_1 \\ = \\ C_3 \leftarrow C_3 - C_1 \end{matrix} \begin{vmatrix} 1 & 0 & 0 \\ 3 & -4 & 1 \\ 1 & 2 & 9 \end{vmatrix}$$

$$= \begin{vmatrix} -4 & 1 \\ 2 & 9 \end{vmatrix} = -36 - 2 = -38$$

$$x = \frac{83}{36} \qquad y = \frac{67}{36} \qquad z = \frac{-38}{36} = \frac{-19}{18}$$

II Metoda Gauss cu substituție descendentă

$$\begin{cases} x + y + 3z = 1 \\ 3x - y + z = 4 & \mid E_2 \leftarrow E_2 - 3E_1 \\ x + 3y - 2z = 10 & \mid E_3 \leftarrow E_3 - E_1 \end{cases}$$

$$\Leftrightarrow \begin{cases} x + y + 3z = 1 \\ -4y - 8z = 1 & \mid : -4 \\ 2y - 5z = 9 \end{cases}$$

$$\begin{cases} x + y + 3z = 1 \\ y + 2z = -\frac{1}{4} \\ 2y - 5z = 9 & \mid E_3 \leftarrow E_3 - 2E_2 \end{cases}$$

$$\begin{cases} x + y + 3z = 1 \\ y + 2z = -\frac{1}{4} \\ -9z = 9 + \frac{1}{2} = \frac{19}{2} \end{cases}$$

$$z = -\frac{19}{18}$$

$$y = {}^{9/}-\frac{1}{4} + {}^{4/}\frac{19}{9} = \frac{76 - 9}{36} = \frac{67}{36}$$

$$x = 1 - \frac{67}{36} + \frac{114}{36} = \frac{47 + 36}{36} = \frac{83}{36}$$

$\square$

ex 3

$$\begin{cases} x + y + 3z = 5 \\ 2x + 2y - 4z = 10 \\ x + y - 7z = 5 \end{cases}$$

Sol :

$$\begin{cases} x + y + 3z = 5 \\ 2x + 2y - 4z = 10 & \mid E_2 \leftarrow E_2 - 2E_1 \\ x + y - 7z = 5 & \mid E_3 \leftarrow E_3 - E_1 \end{cases}$$

$$\Leftrightarrow \begin{cases} x + y + 3z = 5 \\ -10z = 0 \\ -10z = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x + y = 5 \\ z = 0 \end{cases}$$

$S_1 = Sol := \{ (5 - y, \alpha, 0) \mid \alpha \in \mathbb{R} \}$

ex 3'

$$\begin{cases} x + y + 3z = 0 \\ 2x + 2y - 4z = 0 & \mid -2E_1 \\ x + y - 7z = 0 & \mid -E_1 \end{cases}$$

<u>Sol</u>:

$$\begin{cases} x + y + 3z = 0 \\ -10z = 0 \\ -10z = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} x + y = 0 \\ z = 0 \end{cases} \qquad \begin{cases} z = 0 \\ x = -y \end{cases}$$

$$S_2 = Sol = \{ (-y, y, 0) \mid y \in \mathbb{R} \}$$

$$S_1 = Sol := \{ (5 - y, y, 0) \mid \alpha \in \mathbb{R} \}$$

$$S_2 = Sol = \{ (-y, y, 0) \mid y \in \mathbb{R} \}$$

$$S_1 = \{ (5, 0, 0) + \vec{x} \mid x \in S_2 \}$$

Metoda Gauss - Jordan

ex 5

$$\begin{cases} x - y + 3z = 2 \\ 2x + 4y - 5z = 7 \\ 3x + 2y + 8z = 10 \end{cases}$$

Sol:

Metoda 1 : Pivotare parțială

$$\left(\begin{array}{ccc|c} \boxed{1} & -1 & 3 & 2 \\ 2 & 4 & -5 & 7 \\ 3 & 2 & 8 & 10 \end{array}\right) \xrightarrow[L_3 \leftarrow L_3 - 3L_1]{L_2 \leftarrow L_2 - 2L_1}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 0 & \boxed{6} & -11 & 3 \\ 0 & 5 & -1 & 4 \end{array}\right) \longrightarrow$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 0 & \boxed{1} & -\frac{11}{6} & \frac{1}{2} \\ 0 & 5 & -1 & 4 \end{array}\right) \xrightarrow[L_3 \leftarrow L_3 - 5L_2]{L_1 \leftarrow L_1 + L_2}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{7}{6} & \frac{5}{2} \\ 0 & 1 & -\frac{11}{6} & \frac{1}{2} \\ 0 & 0 & \boxed{\frac{49}{6}} & \frac{3}{2} \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{7}{6} & \frac{5}{2} \\ 0 & 1 & -\frac{11}{6} & \frac{1}{2} \\ 0 & 0 & \boxed{1} & \frac{9}{49} \end{array}\right) \xrightarrow[L_2 \leftarrow L_2 + \frac{11}{6} L_3]{L_1 \leftarrow L_1 - \frac{7}{6} L_3}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{16}{7} \\ 0 & 1 & 0 & \frac{9}{49} \\ 0 & 0 & 1 & \frac{41}{49} \end{array}\right)$$

$x = \frac{16}{7} \qquad y = \frac{9}{49} \qquad z = \frac{41}{49}$

Metoda 2: Pivotare totală

$$\left(\begin{array}{ccc|c} \boxed{1} & -1 & 3 & 2 \\ 2 & 4 & -5 & 7 \\ 3 & 2 & 8 & 10 \end{array}\right) \xrightarrow[L_3 \leftarrow L_3 - 3L_1]{L_2 \leftarrow L_2 - 2L_1}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & 3 & 2 \\ 0 & 6 & -11 & 3 \\ 0 & 5 & -1 & 4 \end{array}\right) \longrightarrow$$

$$\left(\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & -11 & 6 & 3 \\ 0 & -1 & 5 & 4 \end{array}\right) \longrightarrow$$

$$\left(\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & -1 & 5 & 4 \\ 0 & -11 & 6 & 3 \end{array}\right) \xrightarrow[L_3 \rightarrow L_3 - 11 L_2]{L_1 \rightarrow L_1 + 3L_2}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 14 & 14 \\ 0 & -1 & 5 & 4 \\ 0 & 0 & -49 & -41 \end{array}\right) \longrightarrow$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 14 & 14 \\ 0 & -1 & 5 & 4 \\ 0 & 0 & 1 & \frac{41}{49} \end{array}\right) \xrightarrow[L_2 \leftarrow L_2 - 5L_3]{L_1 \leftarrow L_1 - 14L_3}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{16}{7} \\ 0 & -1 & 0 & -\frac{9}{49} \\ 0 & 0 & 1 & \frac{41}{49} \end{array}\right)$$

$$\begin{array}{c} \begin{array}{ccc} \color{red}{x} & \color{red}{z} & \color{red}{y} \end{array} \\ \left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{16}{7} \\ 0 & 1 & 0 & \frac{9}{49} \\ 0 & 0 & 1 & \frac{41}{49} \end{array}\right. \end{array}$$

$$x = \frac{16}{7} \qquad y = \frac{9}{49} \qquad z = \frac{41}{49}$$

$\square$

Ce se întâmplă dacă avem un sistem incompatibil și aplicăm Gauss - Jordan ?

ex 6

$$\begin{cases} x - y - 3z = 4 \\ 3x + 3y - z = 7 \\ 5x + y - 7z = 10 \end{cases}$$

Sol :

$$A = \left(\begin{array}{ccc|c} \boxed{1} & -1 & -3 & 4 \\ 3 & 3 & -1 & 7 \\ 5 & 1 & -7 & 10 \end{array}\right) \xrightarrow[L_3 \leftarrow L_3 - 5L_1]{L_2 \leftarrow L_2 - 3L_1}$$

$$\left(\begin{array}{ccc|c} 1 & -1 & -3 & 4 \\ 0 & 6 & 8 & -5 \\ 0 & 6 & 8 & -10 \end{array}\right) \to \text{incompatibil}$$

$$-5 \neq -10$$

$$\left(\begin{array}{ccc|c} 1 & -1 & -3 & 4 \\ 0 & 1 & \frac{4}{3} & -\frac{5}{6} \\ 0 & 1 & \frac{4}{3} & -\frac{5}{3} \end{array}\right) \xrightarrow[L_3 \leftarrow L_3 - L_2]{L_1 \leftarrow L_1 + L_2}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -\frac{5}{3} & \frac{19}{6} \\ 0 & 1 & \frac{4}{3} & -\frac{5}{6} \\ 0 & 0 & \frac{4}{3} & -\frac{5}{3} \end{array}\right) \xrightarrow{L_3 \leftarrow L_3 - L_2}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -\frac{5}{3} & -\frac{19}{6} \\ 0 & 1 & \frac{4}{3} & -\frac{5}{6} \\ 0 & 0 & 0 & -\frac{5}{6} \end{array}\right)$$

$$\neq 0 \Rightarrow \text{incompatibil}$$

ex 7

$$\begin{aligned} x_1 + x_2 - x_3 - x_4 &= 0 \\ 3x_1 + 2x_2 + 2x_3 + 3x_4 &= 0 \\ 4x_1 + 3x_2 + x_3 + 2x_4 &= 0 \\ 7x_1 + 5x_2 + 3x_2 + 5x_4 &= 0 \end{aligned}$$

Sol :

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 3 & 2 & 2 & 3 & 0 \\ 4 & 3 & 1 & 2 & 0 \\ 7 & 5 & 3 & 5 & 0 \end{array}\right) \xrightarrow[\substack{L_3 \leftarrow L_3 - 4L_1 \\ L_4 \leftarrow L_4 - 7L_1}]{L_2 \leftarrow L_2 - 3L_1}$$

$$\begin{pmatrix} 1 & 1 & -1 & -1 \\ 0 & -1 & 5 & 6 \\ 0 & -1 & 5 & 6 \\ 0 & -2 & 10 & 12 \end{pmatrix} \xrightarrow[\substack{L_3 \leftarrow L_3 - L_2 \\ L_4 \leftarrow :2}]{L_2 \leftarrow L_2 - L_3}$$

$$\begin{pmatrix} 1 & 1 & -1 & -1 \\ 0 & -1 & 5 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_1 \leftarrow L_1 + L_2}$$

$$\begin{pmatrix} 1 & 0 & 4 & 5 \\ 0 & -1 & 5 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 4 & 5 \\ 0 & 1 & -5 & -6 \end{pmatrix} \rightarrow$$

$$\rightarrow \left(\begin{array}{cc|cc} 1 & 0 & -4 & -5 \\ 0 & 1 & 5 & 6 \end{array}\right)$$

$$x_1 = -4x_3 - 5x_4$$

$$x_2 = 5x_3 + 6x_4$$

Sol : $\{(-4x_3 - 5x_4,\ 5x_3 + 6x_4,\ x_3,\ x_4) \mid x_3, x_4 \in \mathbb{R}\}$

rang $A = 2$

Sistem liniar $\quad A \cdot X = b$

$$A \in \mathcal{M}_{m,n}(\mathbb{R})$$

$$X \in \mathcal{M}_{n,1}(\mathbb{R})$$

$$b \in \mathcal{M}_{m,1}(\mathbb{R})$$

$\boxed{\text{Nr. de necunoscute secundare} = n - \text{rang } A}$